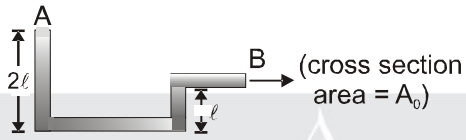
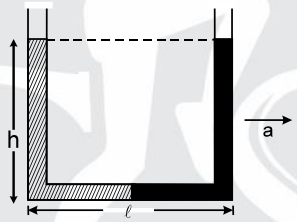
SYLLABUS : FLUID MECHANICS

1. A tube in vertical plane is shown in figure. It is filled with a liquid of density ρ and its end B is closed. Then the force exerted by the fluid on the tube at end B will be : [Neglect atmospheric pressure and assume the radius of the tube to be negligible in comparison to ℓ]



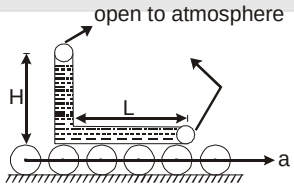
(A) 0 (B) $\rho g \ell A_0$ (C) $2\rho g \ell A_0$ (D) $\frac{\rho g \ell A_0}{2}$

2. A U-tube of base length "ℓ" filled with same volume of two liquids of densities ρ and 2ρ is moving with an acceleration "a" on the horizontal plane as shown in the figure. If the height difference between the two surfaces (open to atmosphere) becomes zero, then the height h is given by:



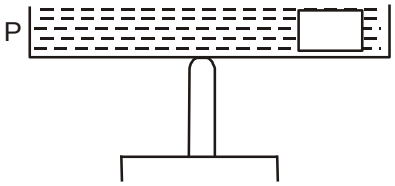
(A) $\frac{a}{2g}\ell$ (B) $\frac{3a}{2g}\ell$ (C) $\frac{a}{g}\ell$ (D) $\frac{2a}{3g}\ell$

3. A narrow tube completely filled with a liquid is lying on a series of cylinders as shown in figure. Assuming no sliding between any surfaces, the value of acceleration of the cylinders for which liquid will not come out of the tube from anywhere is given by



(A) $\frac{gH}{2L}$ (B) $\frac{gH}{L}$ (C) $\frac{2gH}{L}$ (D) $\frac{gH}{\sqrt{2}L}$

4. An open pan P filled with water (density ρ_w) is placed on a vertical rod, maintaining equilibrium. A block of density ρ is placed on one side of the pan as shown in the figure. Water depth is more than height of the block.



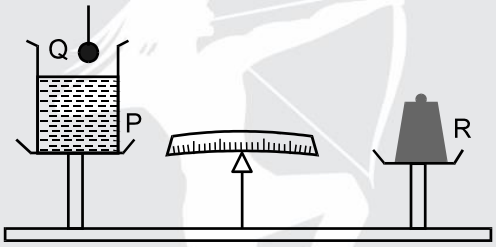
(A) Equilibrium will be maintained only if $\rho < \rho_W$.

(B) Equilibrium will be maintained only if $\rho \leq \rho_W$.

(C) Equilibrium will be maintained for all relations between ρ and ρ_W .

(D) It is not possible to maintained the equilibrium

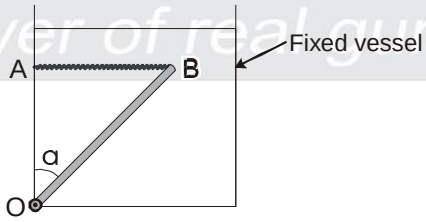
5. Figure shows a weighing-bridge, with a beaker P with water on one pan and a balancing weight R on the other. A solid ball Q is hanging with a thread outside water. It has volume 40 cm^3 and weighs 80 g. If this solid is lowered to sink fully in water, but not touching the beaker anywhere, the balancing weight R' will be



(A) same as R (B) 40 g less than R

(C) 40 g more than R (D) 80 g more than R

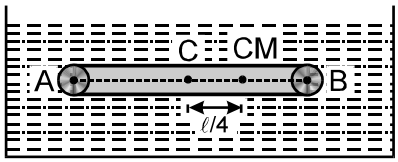
6. A uniform rod OB of length 1m, cross-sectional area 0.012 m^2 and relative density 2.0 is free to rotate about O in vertical plane. The rod is held with a horizontal string AB which can withstand a maximum tension of 45 N. The rod and string system is kept in water as shown in figure. The maximum value of angle α which the rod can make with vertical without breaking the string is



(A) 45º (B) 37º (C) 53º (D) 60º

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7. A non uniform cylinder of mass m, length ℓ and radius r is having its centre of mass at a distance $\ell/4$ from the centre and lying on the axis of the cylinder as shown in the figure. The cylinder is kept in a liquid of uniform density ρ. The moment of inertia of the rod about the centre of mass is I. The angular acceleration of point A relative to point B just after the rod is released from the position shown in figure is



(A) $\frac{\pi\rho g\ell^2 r^2}{I}$ (B) $\frac{\pi\rho g\ell^2 r^2}{4I}$ (C) $\frac{\pi\rho g\ell^2 r^2}{2I}$ (D) $\frac{3\pi\rho g\ell^2 r^2}{4I}$

8. A block of iron is kept at the bottom of a bucket full of water at 2°C. The water exerts buoyant force on the block. If the temperature of water is increased by 1°C the temperature of iron block also increases by 1°C. The buoyant force on the block by water

(A) will increase

(B) will decrease

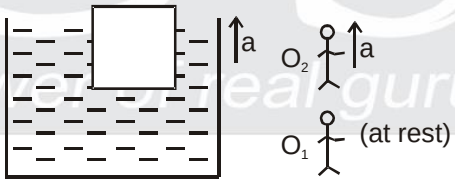
(C) will not change

(D) may decrease or increase depending on the values of their coefficient of expansion

9. A liquid is kept in a cylindrical vessel which is rotated about its axis. The liquid rises at the sides. If the radius of the vessel is 0.05 m and the speed of rotation is 2 rev/s, The difference in the height of the liquid at the centre of the vessel and its sides will be ($\pi^2 = 10$) :

(A) 3 cm (B) 2 cm (C) 3/2 cm (D) 2/3 cm

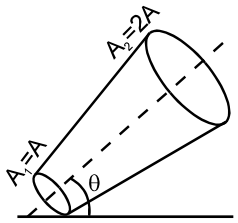
10. A block is partially immersed in a liquid and the vessel is accelerating upwards with an acceleration “a”. The block is observed by two observers O_1 and O_2 , one at rest and the other accelerating with an acceleration “a” upward as shown in the figure. The total buoyant force on the block is :



(A) same for O_1 and O_2 (B) greater for O_1 than O_2

(C) greater for O_2 than O_1 (D) data is not sufficient

11. A portion of a tube is shown in the figure. Fluid is flowing from cross-section area A_1 to A_2. The two cross-sections are at distance 'ℓ' from each other. The velocity of the fluid at section A_2 is $\sqrt{\frac{g\ell}{2}}$. If the pressures at A_1 & A_2 are same, then the angle made by the tube with the horizontal will be:



(A) 37º (B) $\sin^{-1}\frac{3}{4}$ (C) 53º (D) $\cos^{-1}\frac{3}{4}$

12. There is a small hole in the bottom of a fixed container containing a liquid upto height 'h'. The top of the liquid as well as the hole at the bottom are exposed to atmosphere. As the liquid comes out of the hole. (Area of the hole is 'a' and that of the top surface is 'A') :

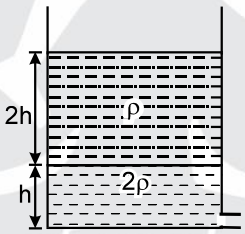
(A) the top surface of the liquid accelerates with acceleration = g

(B) the top surface of the liquid accelerates with acceleration = $g\frac{a^2}{A^2}$

(C) the top surface of the liquid retards with retardation = $g\frac{a}{A}$

(D) the top surface of the liquid retards with retardation = $\frac{ga^2}{A^2}$

13. The velocity of the liquid coming out of a small hole of a large vessel containing two different liquids of densities 2ρ and ρ as shown in figure is



(A) $\sqrt{6gh}$ (B) $2\sqrt{gh}$ (C) $2\sqrt{2gh}$ (D) $\sqrt{gh}$

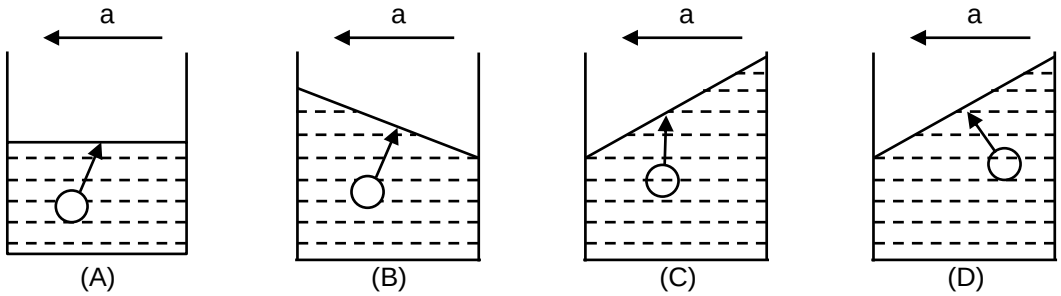
14. Two water pipes P and Q having diameters 2×10^{-2} m and 4×10^{-2} m, respectively, are joined in series with the main supply line of water. The velocity of water flowing in pipe P is

(A) 4 times that of Q (B) 2 times that of Q

(C) 1/2 times that of Q (D) 1/4 times that of Q

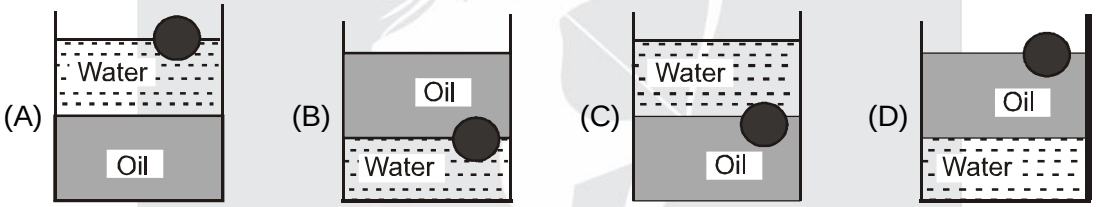
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15. A cup of water is placed in a car moving at a constant acceleration a to the left. Inside the water is a small air bubble. The figure that correctly shows the shape of the water surface and the direction of motion of the air bubble is.



(A) A (B) B (C) C (D) D

16. A ball is made of a material of density ρ where $\rho_{oil} < \rho < \rho_{water}$ with ρ_{oil} and ρ_{water} representing the densities of oil and water, respectively. The oil and water are immiscible. If the above ball is in equilibrium in a mixture of this oil and water, which of the following pictures represents its equilibrium position?



17. Water is flowing continuously from a tap having an internal diameter 8×10^{-3} m. The water velocity as it leaves the tap is 0.4 ms^{-1}. The diameter of the water stream at a distance 2×10^{-1} m below the tap is close to :

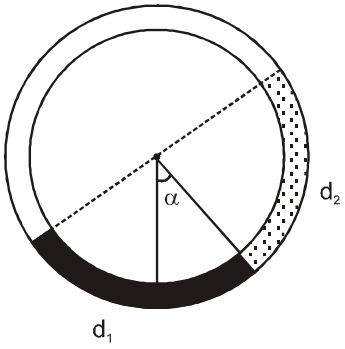
(A) 5.0×10^{-3} m (B) 7.5×10^{-3} m (C) 9.6×10^{-3} m (D) 3.6×10^{-3} m

18. A uniform cylinder of length L and mass M having cross - sectional area A is suspended, with its length vertical, from a fixed point by a massless spring such that it is half submerged in a liquid of density σ at equilibrium position. The extension x_0 of the spring when it is in equilibrium is :

(A) $\frac{Mg}{k}$ (B) $\frac{Mg}{k}\left(1-\frac{LA\sigma}{M}\right)$ (C) $\frac{Mg}{k}\left(1-\frac{LA\sigma}{2M}\right)$ (D) $\frac{Mg}{k}\left(1+\frac{LA\sigma}{M}\right)$

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19. There is a circular tube in a vertical plane. Two liquids which do not mix and of densities d_1 and d_2 are filled in the tube. Each liquid subtands 90° angle at centre. Radius joining their interface makes an angle α with vertical. Ratio $\frac{d_1}{d_2}$ is :



(A) $\frac{1+\sin\alpha}{1-\sin\alpha}$ (B) $\frac{1+\cos\alpha}{1-\cos\alpha}$ (C) $\frac{1+\tan\alpha}{1-\tan\alpha}$ (D) $\frac{1+\sin\alpha}{1-\cos\alpha}$

20. The top of a water tank is open to air and its water level is maintained. It is giving out 0.74 m^3 / min. water per minute through a circular opening of 2 cm radius in its wall. The depth of the centre of the opening from the level of water in the tank is close to :

(A) 2.9 m (B) 9.6 m (C) 4.8 m (D) 6.0 m

21. A liquid of density ρ is coming out of a hose pipe of radius a with horizontal speed v and hits a mesh. 50% of the liquid passes through the mesh unaffected. 25% looses all of its momentum and 25% comes back with the same speed. The resultant pressure on the mesh will be :

(A) $\frac{1}{2}\rho v^2$ (B) $\frac{1}{4}\rho v^2$ (C) $\frac{3}{4}\rho v^2$ (D) ρv^2

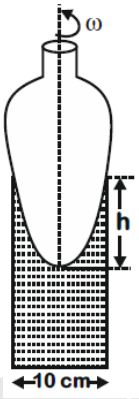
22. A load of mass M kg is suspended from a steel wire of length 2 m and radius 1.0 mm in Searle's apparatus experiment. The increase in length produced in the wire is 4.0 mm. Now the load is fully immersed in a liquid of relative density 2. The relative density of the material of load is 8. The new value of increase in length of the steel wire is :

(A) 3.0 mm (B) zero (C) 5.0 mm (D) 4.0

23. Water flows into a large tank with flat bottom at the rate of 10^{-4} m^3s^{-1}. Water is also leaking out of a hole of area 1 cm2 at its bottom. If the height of the water in the tank remains steady, then this height is

(A) 5.1 cm (B) 1.7 cm (C) 2.9 cm (D) 4 cm

24. A cylindrical vessel containing a liquid is rotated about its axis so that the liquid rises at its sides as shown in the figure. The radius of vessel is 5 cm and the angular speed of rotation is ω rad s^{-1}. The difference in the height, h (in cm) of liquid at the centre of vessel and at the side will be



(A) $\frac{2\omega^2}{25g}$ (B) $\frac{5\omega^2}{2g}$ (C) $\frac{2\omega^2}{5g}$ (D) $\frac{25\omega^2}{2g}$

25. Two identical cylindrical vessels are kept on the ground and each contain the same liquid of density d. The area of the base of both vessels is S but the height of liquid in one vessel is x_1 and in the other, x_2. When both cylinders are connected through a pipe of negligible volume very close to the bottom, the liquid flows from one vessel to the other until it comes to equilibrium at a new height. The change in energy of the system in the process is

(A) $gdS(x_2 + x_1)^2$ (B) $\frac{1}{4}gdS(x_2 - x_1)^2$ (C) $\frac{3}{4}gdS(x_2 - x_1)^2$ (D) $gdS\left(x_2^2 + x_1^2\right)$

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ANSWER KEY

1.	(B)	***2.***	(B)	**3.**	(A)	**4.**	(B)	**5.**	(C)
6.	(B)	**7.**	(B)	**8.**	(A)	**9.**	(B)	**10.**	(A)
11.	(B)	**12.**	(D)	**13.**	(B)	**14.**	(A)	**15.**	(D)
16.	(B)	**17.**	(D)	**18.**	(C)	**19.**	(C)	**20.**	(C)
21.	(C)	**22.**	(A)	**23.**	(A)	**24.**	(D)	**25.**	(B)